



7.1 Suppose that we decompose the schema $R = (A, B, C, D, E)$ into

$$(A, B, C)$$

$$(A, D, E).$$

Show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds:

$$A \rightarrow BC$$

$$CD \rightarrow E$$

$$B \rightarrow D$$

$$E \rightarrow A$$

解:

令 $R_1 = (A, B, C)$, $R_2 = (A, D, E)$, 则 $R_1 \cap R_2 = A$

由 $A \rightarrow BC$ 和 $A \rightarrow A$ 得 $A \rightarrow ABC$, 则 $(R_1 \cap R_2) = A \rightarrow ABC = R_1$

故将 R 分解为 (A, B, C) 和 (A, D, E) 是**无损分解**

7.13 Show that the decomposition in Exercise 7.1 is not a dependency-preserving decomposition.

解:

检验 $B \rightarrow D$, 令 $result = B$, $R_1 = (A, B, C)$, $R_2 = (A, D, E)$

考虑 R_1 : $t = (result \cap R_1)^+ \cap R_1 = B^+ \cap (A, B, C) = (B, D) \cap (A, B, C) = B$,
更新 $result = t \cup B = B$, 无变化

考虑 R_2 : $t = (result \cap R_2)^+ \cap R_2 = \emptyset$, $result$ 仍然无变化

最终 $result$ 不包含 D , 函数依赖 $B \rightarrow D$ 不具有依赖保持性, 故该分解不是保持依赖的分解

7.21 Give a lossless decomposition into BCNF of schema R of Exercise 7.1.

解:

A 和 D 是 R 的超键, $B \rightarrow D$ 不满足 BCNF, 将 R 拆为 $R_1 = B \cup D = (B, D)$ 和 $R_2 = R - (D - B) = (A, B, C, E)$

F 投影到 R_2 有 $F' = \{A \rightarrow BC, E \rightarrow A\}$, 超键为 E , $A \rightarrow BC$ 不满足 BCNF

进一步将 $R_2 = (A, B, C, E)$ 拆除 $R_{21} = (A, B, C)$ 和 $R_{22} = (A, E)$, 均满足 BCNF

综上所述, 有无损 BCNF 分解: $R_1 = B \cup D = (B, D)$, $R_{21} = (A, B, C)$ 和 $R_{22} = (A, E)$

7.22 Give a lossless, dependency-preserving decomposition into 3NF of schema R of Exercise 7.1.

解:

有正则覆盖 $F_c = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

创建关系模式: $R_1 = (A, B, C)$, $R_2 = (C, D, E)$, $R_3 = (B, D)$, $R_4 = (E, A)$

这里 A 是 R 的键, 而 (A, B, C) 包含了 A , 故有无损且保持依赖的 3NF 分解: $R_1 = (A, B, C)$, $R_2 = (C, D, E)$, $R_3 = (B, D)$, $R_4 = (E, A)$

7.30 Consider the following set F of functional dependencies on the relation schema (A, B, C, D, E, G) :

$$A \rightarrow BCD$$

$$BC \rightarrow DE$$

$$B \rightarrow D$$

$$D \rightarrow A$$

- a. Compute B^+ .
- b. Prove (using Armstrong's axioms) that AG is a superkey.
- c. Compute a canonical cover for this set of functional dependencies F ; give each step of your derivation with an explanation.
- d. Give a 3NF decomposition of the given schema based on a canonical cover.
- e. Give a BCNF decomposition of the given schema using the original set F of functional dependencies.

解:

(1)

- 初始化 $result = B$
- 更新 $result = ABCD$ (由 $B \rightarrow D, D \rightarrow A, A \rightarrow BCD$ 推导)
- 更新 $result = ABCDE$ (由 $BC \rightarrow DE$ 推导)

综上所述, $B^+ = \{A, B, C, D, E\}$

(2)

已知 $A \rightarrow BCD$, 由增广律得 $AG \rightarrow BCDG$, 由分解律得 $AG \rightarrow BCG$

已知 $BC \rightarrow DE$, 由增广律得 $BCG \rightarrow DEG$

由传递律得 $AG \rightarrow DEG$, 结合 $AG \rightarrow BCG$ 由合并律得 $AG \rightarrow BCDEG$

又 $AG \rightarrow A$, 所以 $AG \rightarrow ABCDEG$, AG 是超键

(3)

将原依赖集 $F = \{A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A\}$ 右边拆成单属性形式:

$$\{A \rightarrow B, A \rightarrow C, A \rightarrow D, BC \rightarrow D, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$$

删除冗余依赖 $A \rightarrow D$ (可由 $A \rightarrow B, B \rightarrow D$ 推导), 则有:

$$\{A \rightarrow B, A \rightarrow C, BC \rightarrow D, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$$

删除冗余依赖 $BC \rightarrow D$ (可由 $B \rightarrow D$ 推导), 则有:

$$\{A \rightarrow B, A \rightarrow C, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$$

检查 $BC \rightarrow E$ 左边是否有关属性, 考虑 C , 计算 B^+ :

- 初始化 $result = B$
- 更新 $result = ABCD$ (由 $B \rightarrow D, D \rightarrow A, A \rightarrow C$ 推导)
- 更新 $result = ABCDE$ (由 $BC \rightarrow E$ 推导)

故 $B^+ = ABCDE \supseteq E$, 所以 C 是多余属性, 依赖集更新为:

$$\{A \rightarrow B, A \rightarrow C, B \rightarrow D, B \rightarrow E, D \rightarrow A\}.$$

最后把左边相同的依赖合并, 得到正则覆盖:

$$\{A \rightarrow BC, B \rightarrow DE, D \rightarrow A\}$$

(4)

利用正则覆盖拆分得到: $R_1 = (A, B, C), R_2 = (B, D, E), R_3 = (D, A)$

由于无依赖可推出 G , 所以候选键必须包含 G , 当前子模式均不包含候选键, 由第 (2) 问得 AG 是超键, 可以新增 $R_4 = (A, G)$

综上所述有 3NF 分解: $(A, B, C), (B, D, E), (D, A), (A, G)$

(5)

已知函数依赖集等价于 $\{A \rightarrow BC, B \rightarrow DE, D \rightarrow A\}$

A 不是 R 的超键, $A \rightarrow BC$ 违反 BCNF, 将 R 分解为 $R_1 = (A, B, C)$ 和 $R_2 = (A, D, E, G)$

考虑 R_1 , 满足 BCNF

考虑 R_2 , D 不是 R_2 的超键, $D \rightarrow A$ 违反 BCNF, 将 R_2 分解为 $R_{21} = (A, D)$ 和 $R_{22} = (D, E, G)$, 满足 BCNF

综上所述有 BCNF 分解:

$$(A, B, C), (A, D), (D, E, G)$$

7.32 Consider the schema $R = (A, B, C, D, E, G)$ and the set F of functional dependencies:

$$A \rightarrow BC$$

$$BD \rightarrow E$$

$$CD \rightarrow AB$$

- a. Find a nontrivial functional dependency containing no extraneous attributes that is logically implied by the above three dependencies and explain how you found it.**
- b. Use the BCNF decomposition algorithm to find a BCNF decomposition of R . Start with $A \rightarrow BC$. Explain your steps.**
- c. For your decomposition, state whether it is lossless and explain why.**
- d. For your decomposition, state whether it is dependency preserving and explain why.**

解:

(1)

将原依赖集 $F = \{A \rightarrow BC, BD \rightarrow E, CD \rightarrow AB\}$ 右边拆成单属性形式:

$$\{A \rightarrow B, A \rightarrow C, BD \rightarrow E, CD \rightarrow A, CD \rightarrow B\}$$

删除冗余依赖 $CD \rightarrow B$ (可由 $CD \rightarrow A, A \rightarrow B$ 推导), 则有:

$$\{A \rightarrow B, A \rightarrow C, BD \rightarrow E, CD \rightarrow A\}$$

检查后发现无冗余属性, 故有正则覆盖:

$$\{A \rightarrow BC, BD \rightarrow E, CD \rightarrow A\}$$

综上所述, 有满足要求的非平凡依赖 $A \rightarrow B$

(2)

A 不是 R 的超键, $A \rightarrow BC$ 违反 BCNF, 将 R 分解为 $R_1 = (A, B, C)$ 和 $R_2 = (A, D, E, G)$

在 R_2 中, 由 $A \rightarrow BC$ 和 $BD \rightarrow E$ 得 $AD \rightarrow BD \rightarrow E$ 即 $AD \rightarrow E$, 而 AD 不是 R_2 的超键, 违反 BCNF, 将 R_2 分解为 $R_{21} = (A, D, E)$ 和 $R_{22} = (A, D, G)$, 均满足 BCNF

综上所述, 有 BCNF 分解:

$$(A, B, C), (A, D, E), (A, D, G)$$

(3)

BCNF 分解算法每一步都严格按照无损规则拆分, 所以是无损分解。

(4)

已知 $R_1 = (A, B, C)$, $R_{21} = (A, D, E)$ $R_{22} = (A, D, G)$, 考虑 $BD \rightarrow E$:

- 初始化 $result = BD$
- 考虑 R_1 : $t = (result \cap R_1)^+ \cap R_1 = B^+ \cap (A, B, C) = B$, $result$ 无变化
- 考虑 R_{21} : $t = (result \cap R_{21})^+ \cap R_{21} = D^+ \cap (A, D, E) = D$, $result$ 无变化
- 考虑 R_{22} : $t = (result \cap R_{22})^+ \cap R_{22} = D^+ \cap (A, D, E) = D$, $result$ 无变化

最终 $result$ 不包含 E , 函数依赖 $BD \rightarrow E$ 不具有依赖保持性, 故该分解不是保持依赖的分解